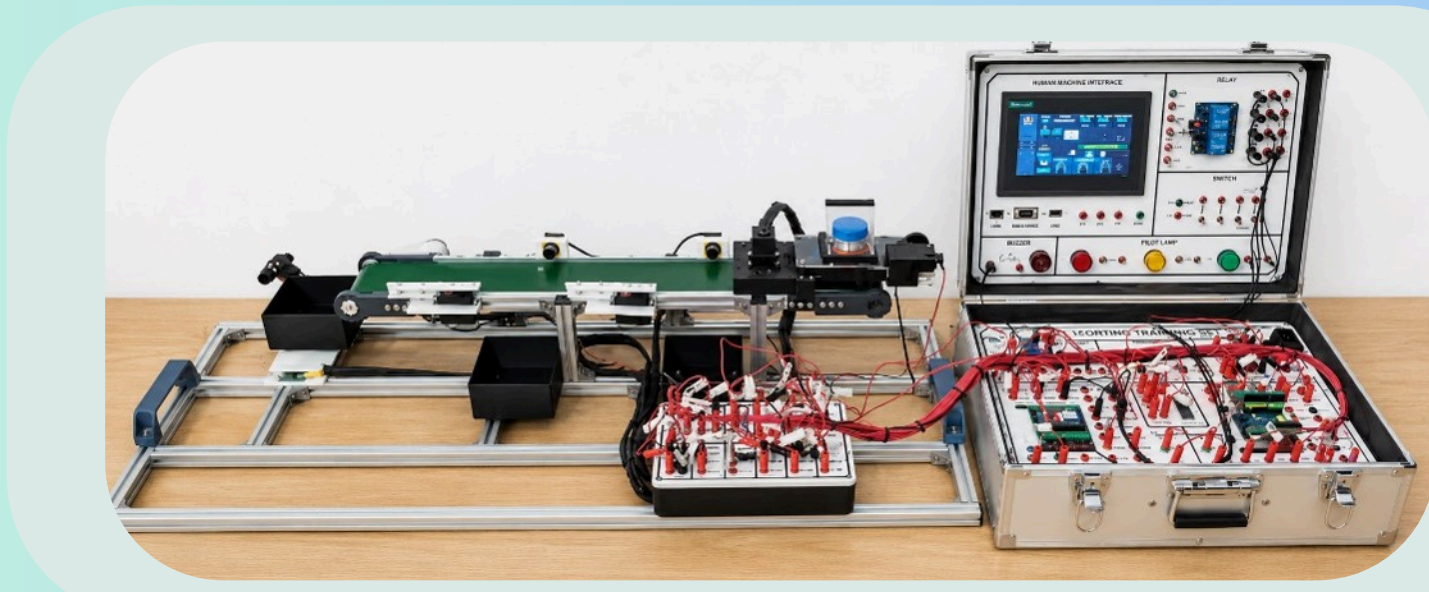


Design and Development of an Internet of Things-Based Sorting Training Set as a Learning

Introduction

- The era of digital transformation and IoT demands vocational learning media capable of integrating control systems, data processing, and real-time monitoring.
- Current practicum media in the laboratory lack integration between mechanical control and internet connectivity, limiting students' understanding of industrial IoT ecosystems.
- This project develops an IoT-based Sorting Training Set utilizing a dual-microcontroller architecture (Arduino and ESP32) and an HMI for automated sorting simulation and real-time data visualization.

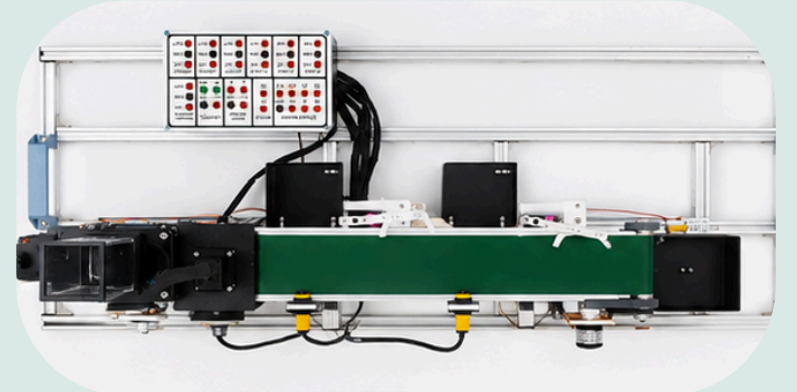


Hardware System

The system is housed in a compact portable box containing the main controller (PLC/Microcontroller), motor drivers for the 75cm conveyor, proximity sensors, relays, and IoT communication modules to ensure seamless sorting operations.

How The System Work

Objects are placed onto the 75cm conveyor actuator. As they move, sensors integrated into the portable trainer box detect the object's properties (e.g., material or height). The control unit processes this data to trigger the sorting mechanism, pushing objects into their designated categories while sending real-time data to the IoT dashboard.



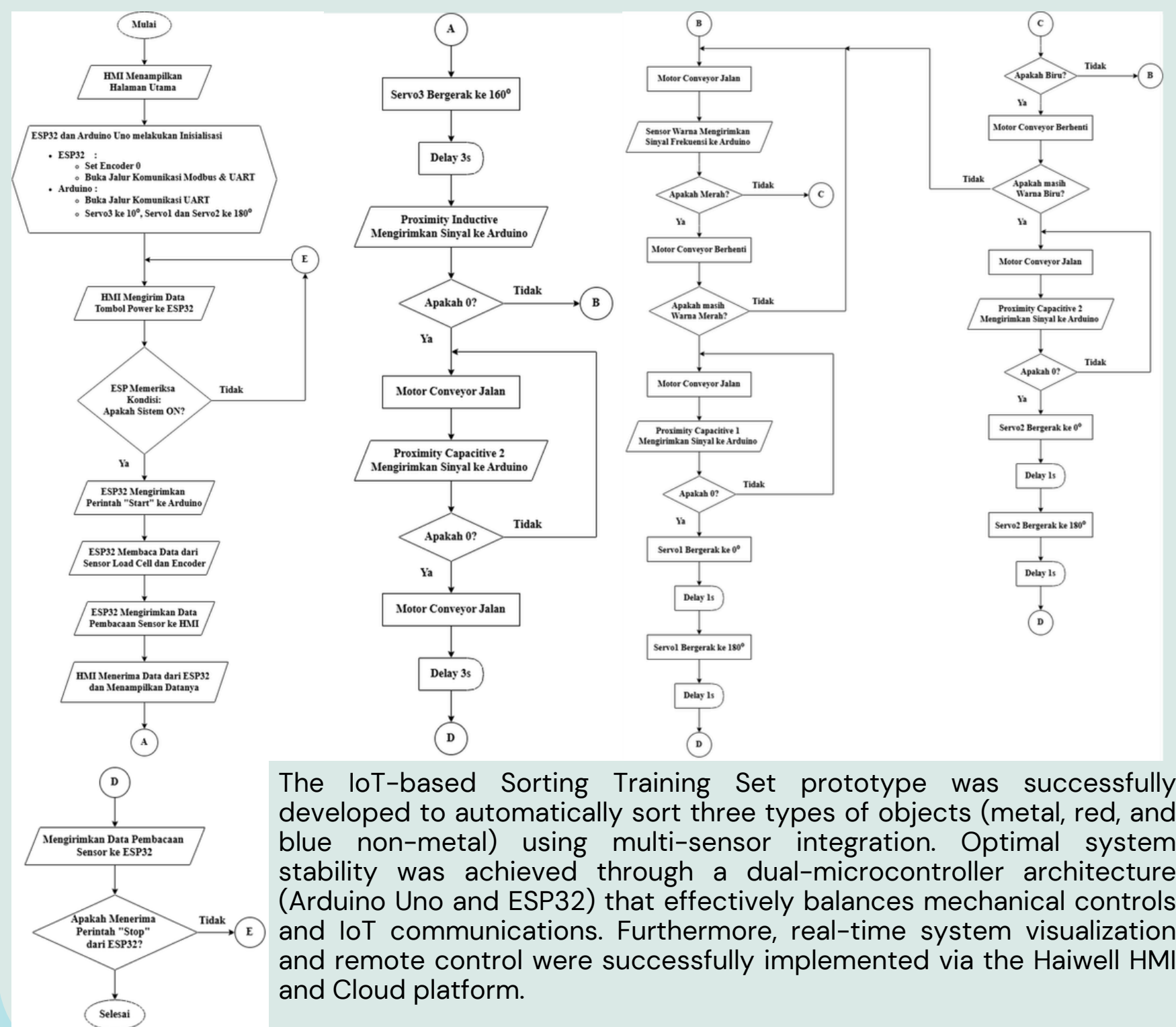
Objectives

- To develop an IoT-based automated Sorting Training Set prototype on a mini conveyor as an interactive laboratory learning media.
- To implement a dual-microcontroller architecture (Arduino Uno & ESP32) to balance the workload between mechanical control and network communication.
- To provide a real-time object sorting monitoring system via a Human-Machine Interface (HMI) screen.

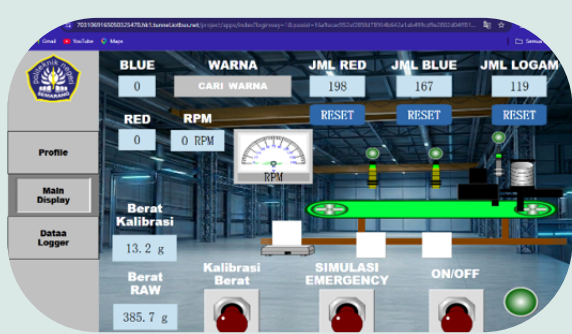
Conclusion

The IoT-based Sorting Training Set prototype was successfully developed to automatically sort three types of objects (metal, red, and blue non-metal) using multi-sensor integration. Optimal system stability was achieved through a dual-microcontroller architecture (Arduino Uno and ESP32) that effectively balances mechanical controls and IoT communications. Furthermore, real-time system visualization and remote control were successfully implemented via the Haiwell HMI and Cloud platform.

Flowchart



HMI Design



This interactive HMI dashboard serves as the primary interface for controlling and continuously monitoring the automated sorting system in real time, displaying live material counts, motor RPM, and weight measurements.



Bintang Briliansyah
(3.32.23.0.06)



Habib Husein Baydawi
(3.32.23.0.11)



Khusna Khasanatur Rovi'a
(3.32.23.0.12)



Supervisor 1
Ilham Sayekti, S.T., M.Kom.



Supervisor 2
Muhamad Cahyo Ardi Prabowo, S.T., M.Tr.T.